Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/02/2025
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	Plant Water Management 5 and 6		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mmw@cecs.pdx.edu	**Fax:**	FY 503-725-8255
PI Organization Type:	UNIVERSITY	**Phone:**	503 725-4292
Organization Name:	Portland State University		
PI Address 1:	MCECS Research		
PI Address 2:	P.O. Box 751		
PI Web Page:	https://		
City:	Portland	**State:**	OR
Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Flight	**Solicitation / Funding Source:**	Small Business Innovative Research (SBIR) Phase III
Start Date:	04/13/2022	**End Date:**	04/12/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: The organizational affiliation for this SBIR investigation is IRPI, LLC.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Contract 80NSSC22C0012		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The International Space Station (ISS) Program Division of the Science Mission Directorate (SMD) is seeking to fund work related to plant water management research. The research seeks to explore the fundamental mechanisms and the design space for an autonomous passive farming system in space.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	This investigation continues as "Plant Water Management 5 and 6" (Contract #80NSSC23CA033) under same PI: Mark Weislogel, Ph.D. See that project for updated reporting. January 2025 update, compiled from information submitted by NASA Glenn Research Center: The NASA Plant Water Management (PWM) technology demonstrations aboard the International Space Station (ISS) apply recent advances in microgravity capillary fluidics research towards the challenges of watering plants in space. Plant growth in a low-g environment is often hampered by inadequate aeration and oversaturation of the root zone. This investigation has aimed to exploit the passive capillary forces of poorly wetting liquids (i.e., contaminated water) within unique system geometries that effectively replace the role of gravity in providing sufficient aeration and hydration for simulated plants. Several flight demonstrations have been completed on the ISS, including soil and hydroponic models in single and parallel channel networks. The results to date demonstrate proof-of-concept, system stability, limits of operation, and more. The implications are discussed in relation to plant growth facilities for further near-term microgravity plant science research, as well as for automated food production for long duration human exploration missions. [Ed. Note: For additional information, see the published papers in the Cumulative Bibliography.]
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Papers from Meeting Proceedings	Hatch T, Wasserman M, McQuillen J, Weislogel M. "The Plant Water Management experiments: Soil." 51st International Conference on Environmental Systems, Saint Paul, MN, July 10-14, 2022. 51st International Conference on Environmental Systems, Saint Paul, MN, July 10-14, 2022. ICES paper ICES-2022-013. , Jul-2022
Papers from Meeting Proceedings	Hatch T, Wasserman M, McQuillen J, Weislogel M. "Plant water management in microgravity." 51st International Conference on Environmental Systems, Saint Paul, MN, July 10-14, 2022. 51st International Conference on Environmental Systems, Saint Paul, MN, July 10-14, 2022. ICES paper ICES-2022-011. , Jul-2022